Math709 – Foundations of Computational Mathematics II

Spring 2013

Midterm Exam

You must show all of your work to get a credit for a correct answer.

1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

   (a) Find a full SVD for $\mathbf{A}$.

   (b) Compute $\|\mathbf{A}\|_1$, $\|\mathbf{A}\|_2$, $\|\mathbf{A}\|_\infty$, $\|\mathbf{A}\|_F$.

2. (20 points) Let $\mathbf{A} \in \mathbb{C}^{m\times m}$.

   (a) Under what conditions is the QR factorization unique if $\mathbf{A}$ is full rank?

   (b) Write out a pseudo code for the QR factorization using modified Gram-Schmidt orthogonolization.

3. (20 points)

   (a) Suppose that $\mathbf{x}, \mathbf{y} \in \mathbb{C}^m$ are unit vectors, i.e., $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 = 1$. Construct a unitary matrix $\mathbf{F}$ such that $\mathbf{Fx} = \mathbf{y}$.

   (b) For general $\mathbf{x}, \mathbf{y} \in \mathbb{C}^m$, can we always find such a unitary matrix $\mathbf{F}$ such that $\mathbf{Fx} = \mathbf{y}$? If no, what is the requirement on $\mathbf{x}, \mathbf{y}$?

4. (20 points) Given $\mathbf{A} \in \mathbb{C}^{m\times n}$ of full rank n and $b \in \mathbb{C}^m$, consider the block 2×2 system of equations

$$\begin{bmatrix} \mathbf{I} & \mathbf{A} \\ \mathbf{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{r} \\ \mathbf{x} \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}.$$

   Show that this system has a unique solution $(\mathbf{r}, \mathbf{x})^T$ and that the vectors $\mathbf{r}$ and $\mathbf{x}$ are the residual and the solution of the least squares problem $\min_{\mathbf{x}\in\mathbb{C}^n} \|\mathbf{Ax} - \mathbf{b}\|_2$.

5. (20 points)

   (a) Define the three terms "condition of a problem", "backwards stability of an algorithm" and "accuracy of an algorithm".

   (b) Prove that a backwards stable algorithm applied to a problem with condition number κ is *accurate* and provide an error estimate in terms of κ.

6. (20 points) Let $\mathbf{A} \in \mathbb{C}^{m\times m}$. Suppose for each k with $1 \le k \le m$, the upper-left $k \times k$ block $\mathbf{A}_{1:k,1:k}$ is nonsingular.

   (a) Show that $\mathbf{A}$ has a LU factorization. (*Hint: use the method of induction*)

   (b) What can you say about the pattern of $\mathbf{L}$ and $\mathbf{U}$ if $\mathbf{A}$ is banded with bandwidth $2p+1$, i.e., $a_{ij} = 0$ for $|i-j| > p$?